

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 02: CELL: THE BUILDING BLOCK OF LIFE

TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 Which organelle forms the primary site of ribosome subunit synthesis inside the eukaryotic cell?

- | | |
|---------------------|---------------------------|
| (A) Centrosome | (B) Nucleolus |
| (C) Golgi apparatus | (D) Endoplasmic reticulum |

Q.2 What is the biochemical nature of the middle lamella that cements adjacent plant cell walls together?

- | | |
|-----------------------------------|------------------------|
| (A) Calcium and Magnesium pectate | (B) Lignin and suberin |
| (C) Pure cellulose microfibrils | (D) Starch granules |

Q.3 Which cellular transport mechanism consumes metabolic ATP energy to pump ions against an electrochemical concentration gradient?

- | | |
|---------------------------|----------------------|
| (A) Facilitated diffusion | (B) Active transport |
| (C) Simple diffusion | (D) Osmosis |

Q.4 What unique biochemical feature distinguishes the mitochondrial inner membrane from other eukaryotic membranes?

- | | |
|-------------------------------------|---|
| (A) It contains cellulose | (B) It has an unusually high protein-to-lipid ratio (around 75% protein) and contains cardiolipin |
| (C) It lacks any transport proteins | (D) It is completely impermeable to water |

Q.5 When ripe tomatoes turn from green to bright red during ripening, which organellar transformation occurs?

- | | |
|---|---|
| (A) Leucoplasts transform into chloroplasts | (B) Chloroplasts transform into chromoplasts as chlorophyll breaks down and lycopene carotenoids accumulate |
| (C) Vacuoles turn into mitochondria | (D) Ribosomes synthesize hemoglobin |

Q.6 Which cytoplasmic organelle is directly responsible for organizing mitotic spindle microtubules in animal cells?

- | | |
|-----------------------------|---------------|
| (A) Centrosome (Centrioles) | (B) Lysosome |
| (C) Peroxisome | (D) Nucleolus |

Q.7 Which specialized microbody in plant cells converts stored fatty acids into carbohydrates during seed germination?

- | | |
|----------------|--------------|
| (A) Glyoxysome | (B) Lysosome |
| (C) Centriole | (D) Ribosome |
-

Q.8 What is the primary function of plasmodesmata in plant tissues?

- | | |
|---------------------------|---|
| (A) To prevent water loss | (B) To provide microscopic cytoplasmic channels connecting adjacent plant cells for intercellular transport and signaling |
| (C) To synthesize starch | (D) To anchor roots to soil |
-

Q.9 Which organelle acts as a dynamic intracellular calcium (Ca^{2+}) reservoir in skeletal muscle fibers?

- | | |
|---|------------------|
| (A) Sarcoplasmic Reticulum (modified Smooth ER) | (B) Mitochondria |
| (C) Lysosomes | (D) Nucleus |
-

Q.10 Which eukaryotic organelle lacks a surrounding lipid membrane entirely?

- | | |
|----------------|--------------|
| (A) Peroxisome | (B) Ribosome |
| (C) Lysosome | (D) Vacuole |
-

Q.11 Why does a plant cell not undergo cytolysis (bursting) when placed in a hypotonic medium where internal turgor pressure reaches 15 atmospheres?

- | | |
|-----------------------------------|---|
| (A) Water stops entering at 1 atm | (B) The cellulose cell wall exerts an equal inward mechanical wall pressure, maintaining structural integrity |
| (C) Cell sap is non-osmotic | (D) The tonoplast dissolves |
-

Q.12 What constitutes the 'nucleoid' of a bacterium?

- | | |
|--|---|
| (A) A membrane-bound nucleus with multiple chromosomes | (B) A single circular double-stranded DNA molecule without a nuclear envelope or histone proteins |
| (C) A clump of ribosomes | (D) A protein capsule |
-

Q.13 Which organelle contains hydrolytic enzymes that function optimally at an acidic pH of ~4.5 to 5.0?

- | | |
|--------------------------|------------------------|
| (A) Lysosome | (B) Peroxisome |
| (C) Mitochondrial matrix | (D) Chloroplast stroma |
-

Q.14 The stacks of flattened disc-like thylakoid sacs inside a chloroplast are termed:

- | | |
|---------------|------------|
| (A) Cristae | (B) Grana |
| (C) Cisternae | (D) Stroma |
-

Q.15 The trans-face of the Golgi apparatus is also known as the:

- | | |
|------------------|--------------------------|
| (A) Forming face | (B) Maturing / Exit face |
| (C) Inner face | (D) Entry cis-face |
-

Q.16 Which antibiotic inhibits bacterial cell wall synthesis by blocking peptidoglycan cross-linking?

- | | |
|----------------|-------------|
| (A) Penicillin | (B) Insulin |
| (C) Adrenaline | (D) Aspirin |
-

Q.17 Which cellular organelle is responsible for decomposing toxic hydrogen peroxide (H_2O_2) into water and oxygen using catalase?

- | | |
|---------------------|----------------|
| (A) Peroxisome | (B) Ribosome |
| (C) Golgi apparatus | (D) Centrosome |
-

Q.18 A cell that has lost its nucleus, ribosomes, and mitochondria yet remains metabolically active in vascular transport for years is the:

- (A) Sieve tube element in phloem (B) Xylem vessel
(C) Tracheid (D) Guard cell

Q.19 The proteinaceous core structure that holds sister chromatids together at the centromere and binds spindle microtubules is the:

- (A) Kinetochore (B) Centrosome
(C) Telomere (D) Chromomere

Q.20 Which type of cell division produces four non-identical haploid gametes from a diploid progenitor cell?

- (A) Mitosis (B) Meiosis
(C) Amitosis (D) Binary fission

Q.21 What happens when a cell's lysosomes rupture en masse under pathological stress?

- (A) The cell synthesizes more ATP (B) Autolysis (self-digestion) occurs, leading to programmed necrosis of the cell
(C) The cell becomes immortal (D) The cell wall thickens

Q.22 What is the structural diameter of a eukaryotic 80S ribosome?

- (A) 20–30 nm (B) 2 μ m
(C) 200 μ m (D) 1 mm

Q.23 Which of the following cellular structures is present in prokaryotes, fungi, and plants, but entirely absent in animals?

- (A) Mitochondria (B) Cell wall
(C) Plasma membrane (D) Ribosomes

Q.24 The liquid ground substance filling the chloroplast interior surrounding thylakoids is the:

- (A) Matrix (B) Stroma
(C) Cytosol (D) Tonoplasm

Q.25 Which specialized lipid molecule stabilizes plasma membrane fluidity across varying temperatures in mammalian cells?

- (A) Cellulose (B) Cholesterol
(C) Triglyceride (D) Starch

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(B)	The nucleolus is the specialized nuclear sub-compartment where ribosomal RNA (rRNA) is transcribed and combined with ribosomal proteins.
Q.2	(A)	The middle lamella is an extracellular cementing layer composed of sticky calcium and magnesium pectate.
Q.3	(B)	Active transport moves ions against concentration gradients using membrane protein pumps powered by ATP hydrolysis.
Q.4	(B)	The inner mitochondrial membrane is packed with respiratory chain enzyme complexes and ATP synthases, giving it a 3:1 protein-to-lipid ratio.
Q.5	(B)	Ripening involves chloroplast degradation and synthesis of red/orange carotenoid pigments within chromoplasts.
Q.6	(A)	Centrosomes contain a pair of centrioles that serve as the main microtubule-organizing centers (MTOCs) during animal cell mitosis.
Q.7	(A)	Glyoxysomes are specialized plant peroxisomes that run the glyoxylate cycle to convert stored lipids into sucrose.
Q.8	(B)	Plasmodesmata are membrane-lined trans-wall pores that maintain cytoplasmic continuity (symplast) across adjacent plant cells.
Q.9	(A)	The sarcoplasmic reticulum is specialized smooth ER that sequesters and releases Ca^{2+} ions to regulate muscle contraction.
Q.10	(B)	Ribosomes are non-membrane-bound ribonucleoprotein complexes found in both prokaryotes and eukaryotes.
Q.11	(B)	The tensile strength of the cellulosic cell wall provides counter wall pressure ($\text{\$WP = TP\$}$), preventing osmotic lysis.
Q.12	(B)	The bacterial nucleoid is a region containing naked, uncompartimentalized, circular genomic DNA.
Q.13	(A)	Lysosomal acid hydrolases require an acidic luminal pH maintained by proton-pumping v-type ATPases.
Q.14	(B)	Thylakoids are organized into membranous stacks called grana (singular: granum), where light reactions occur.
Q.15	(B)	Cis-face is the receiving/forming face facing the ER; trans-face is the exit/maturing face that pinches off secretory vesicles.
Q.16	(A)	Penicillin targets transpeptidase enzymes that cross-link peptidoglycan in bacterial cell walls, causing bacterial lysis without harming human cells.
Q.17	(A)	Peroxisomes contain catalase and oxidases to detoxify hydrogen peroxide and break down long-chain fatty acids.
Q.18	(A)	Phloem sieve tube elements lack nuclei at maturity but stay alive through metabolic support from companion cells.
Q.19	(A)	Kinetochores are disc-shaped protein complexes assembled on centromeric DNA where spindle fibers attach.
Q.20	(B)	Meiosis is reduction division consisting of two successive nuclear divisions yielding four haploid gametes.
Q.21	(B)	Widespread lysosomal leakage releases acid hydrolases into the neutral cytosol, digesting cellular components (autolysis).
Q.22	(A)	Ribosomes are nanometer-scale complexes measuring roughly 20 to 30 nanometers in diameter.
Q.23	(B)	Cell walls occur in bacteria (peptidoglycan), fungi (chitin), and plants (cellulose), but never in animal cells.
Q.24	(B)	The stroma contains soluble enzymes (e.g., RuBisCO) that catalyze the dark reactions of the Calvin cycle.
Q.25	(B)	Cholesterol intercalates between phospholipids, preventing membrane crystallization at low temperatures and excessive fluidity at high temperatures.